

Singularities

A point at which $f(z)$ is not analytic is called singularity or singular point.

Zeros of Function

Points at which $f(z) = 0$.

Types of Singularities

Laurent series of a function $f(z)$,

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n}$$

Here, $\sum b_n (z-a)^{-n}$ is called principle part of Laurent's series.

1. Removable Singularity:

$z=a$ is called removable singularity if there are zero number of terms in principle part.

$$\text{i.e. } f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n.$$

Note: ① No. of times of pole is order of pole.

② Pole of order 1 is called simple pole.

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2. Poles Singularity:

If there are finite number of terms in principle part, then $z=a$ is called poles.

$$f(z) = \sum a_n (z-a)^n + \sum_{n=1}^m b_n (z-a)^{-n}$$

$$\Rightarrow f(z) = \sum a_n (z-a)^n + \left[\frac{b_1}{(z-a)} + \frac{b_2}{(z-a)^2} + \dots + \frac{b_m}{(z-a)^m} \right]$$

We can find poles by $z-a=0 \Rightarrow z=a$.

3. Essential Singularity:

If there are infinite number of terms in principle part then $z=a$ is called essential singularity.

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n}$$

(i) Isolated Essential Singularity:

Limiting points of zeroes of function are called isolated essential singularity.

(ii) Non-isolated Essential Singularity:

Limiting points of poles are called non-isolated essential singularity.

Expansions:

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$$

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots$$

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$$

Q.1) Find singularities of the function,

$$f(z) = \frac{\sin z}{z}$$

Solⁿ:

$$f(z) = \frac{\left[z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots \right]}{z}$$

$$\Rightarrow f(z) = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \dots$$

$\therefore$ There is no term of negative power of z .

$\therefore$ Principle part is 0.

So, $z=0$ is Removable Singularity.

Q.2) Find the singularity of $f(z) = e^{1/z}$.

Solⁿ:

$$f(z) = 1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots$$

$\therefore$ There are infinite number of terms in principle part.

$\therefore z=0$ is essential singularity.

Zeros of $f(z)$ are $e^{1/z} = 0$

$$\Rightarrow \frac{1}{z} = -\infty \quad \left[e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0 \right]$$

$$\Rightarrow z = 0$$

$z=0$, which is limiting point of zero.

So, $z=0$ is isolated essential singularity.

Q.3) Find the singularity of $f(z) = \frac{1}{(z-5)^3 (z-4)^2}$.

Solⁿ:

For poles,

$$(z-5)^3 (z-4)^2 = 0$$

$$\Rightarrow z = 5, 4$$

Here, 5 is a pole of order 3 and 4 is a pole of order 2.

Q.4) What kind of singularity have the following function,

$$f(z) = \frac{1 - e^z}{1 + e^z} \text{ at } z = \infty.$$

Solⁿ:

$$\text{For poles, } 1 + e^z = 0$$

$$\Rightarrow e^z = -1$$

We have,

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\text{For } \theta = (2n+1)\pi,$$

$$e^{i(2n+1)\pi} = \cos(2n+1)\pi + i \sin(2n+1)\pi$$

$$\Rightarrow e^{i(2n+1)\pi} = -1 + 0 = -1$$

$$\text{So, } e^z = e^{(2n+1)\pi i}$$

$$\Rightarrow z = (2n+1)\pi i$$

$$\Rightarrow z = \pi i, 3\pi i, 5\pi i, \dots, \infty$$

$\therefore z = \infty$ is limiting ^{point} of poles.

$\therefore z = \infty$ is non-isolated essential singularity.

Q.5) Discuss the singularity of function $f(z) = \frac{e^{1/z}}{z^2}$.

Solⁿ:

$$f(z) = \frac{e^{1/z}}{z^2}$$

$$\Rightarrow f(z) = \frac{1}{z^2} \left[1 + \frac{1}{z} + \frac{1}{2! z^2} + \frac{1}{3! z^3} + \dots \right]$$

$$\Rightarrow f(z) = \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{2! z^4} + \frac{1}{3! z^5} + \dots$$

$\therefore z = 0$ is limiting point of poles of $f(z)$.

$\therefore$ There are infinite number of terms in principle part.

So, $z = 0$ is non-isolated essential singularity.

Q.6) Discuss the singularity of $f(z) = z \operatorname{cosec} z$ at $z = \infty$.

Solⁿ:

$$f(z) = z \operatorname{cosec} z$$

$$\Rightarrow f(z) = \frac{z}{\sin z}$$

For poles, $\sin z = 0$

$$\Rightarrow z = n\pi$$

$$\Rightarrow z = 0, \pi, 2\pi, 3\pi, \dots, \infty$$

$\therefore z = \infty$ is limiting point of poles.

$\therefore z = \infty$ is non-isolated essential singularity.

Q.7) Discuss the singularity of $f(z) = \frac{1 - \cos z}{z}$.

Solⁿ:

$$f(z) = \frac{1 - \cos z}{z}$$

$$\Rightarrow f(z) = \frac{1}{z} \left[1 - \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots \right) \right]$$

$$\Rightarrow f(z) = \frac{1}{z} \left[\frac{z^2}{2!} - \frac{z^4}{4!} + \dots \right]$$

$$\Rightarrow f(z) = \frac{z}{2!} - \frac{z^3}{4!} + \dots$$

$\therefore$ There is no term of negative power of z .

$\therefore$ Principle part is 0.

So, $z=0$ is Removable singularity.

Q.8) Discuss the singularity of $f(z) = \frac{1}{\sin z - \cos z}$.

Solⁿ:

For poles, $\sin z - \cos z = 0$

$$\Rightarrow \sqrt{2} \left[\frac{1}{\sqrt{2}} \sin z - \frac{1}{\sqrt{2}} \cos z \right] = 0$$

$$\Rightarrow \sqrt{2} \left[\cos \frac{\pi}{4} \cdot \sin z - \sin \frac{\pi}{4} \cdot \cos z \right] = 0$$

$$\Rightarrow \sqrt{2} \cdot \sin \left(z - \frac{\pi}{4} \right) = 0$$

$$\Rightarrow \sin \left(z - \frac{\pi}{4} \right) = 0$$

$$\Rightarrow z - \frac{\pi}{4} = n\pi \quad [\sin n\pi = 0]$$

$$\Rightarrow z = n\pi + \frac{\pi}{4}$$

$$\Rightarrow z = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4} \dots$$

$\therefore z = \frac{\pi}{4}, \frac{5\pi}{4}, \dots$ are poles of $f(z)$.

Q.9) Find the kind of singularity of $f(z) = \tan\left(\frac{1}{z}\right)$ at $z = 0$?

Solⁿ:

$$f(z) = \tan\left(\frac{1}{z}\right)$$

$$\Rightarrow f(z) = \frac{\sin 1/z}{\cos 1/z}$$

For poles, $\cos 1/z = 0$

$$\Rightarrow \frac{1}{z} = (2n \pm 1) \frac{\pi}{2}$$

$$\Rightarrow z = \frac{1}{(2n \pm 1) \pi/2}$$

$$\Rightarrow z = \frac{2}{\pi}, -\frac{2}{\pi}, \frac{2}{3\pi} \dots \dots 0$$

$\therefore z = 0$ is limiting point of pole.

$\therefore z = 0$ is non-isolated essential singularity.

Q.10) Show that $f(z) = e^{-1/z^2}$ has no singularity.

Solⁿ:

$$f(z) = \frac{1}{e^{1/z^2}}$$

Since, $e^{1/z^2} \neq 0$ for any value of z .

$\therefore f(z)$ has no singularity.

Q.11) Discuss the singularity of $f(z) = ze^{1/z^2}$ at $z = \infty$.

Solⁿ:

$$f(z) = ze^{1/z^2}$$

Rough

$$\left\{ \begin{aligned} \Rightarrow f(z) &= z \left[1 + \frac{1}{z^2} + \frac{1}{2!z^4} + \frac{1}{3!z^6} + \dots \right] \\ \Rightarrow f(z) &= z + \frac{1}{z} + \frac{1}{2!z^3} + \frac{1}{3!z^5} + \dots \end{aligned} \right\}$$

$$\text{Let, } z = \frac{1}{\omega}$$

$$\Rightarrow f\left(\frac{1}{\omega}\right) = \frac{e^{\omega^2}}{\omega} = \frac{\left(1 + \omega^2 + \frac{\omega^4}{2!} + \frac{\omega^6}{3!} + \dots\right)}{\omega}$$

$$\Rightarrow f\left(\frac{1}{\omega}\right) = \frac{1}{\omega} + \omega + \frac{\omega^3}{2!} + \frac{\omega^5}{3!} + \dots$$

$\therefore$ There is only one term of negative power of ω .

$\therefore \omega = 0$ is pole of $f(1/\omega)$.

We have, $z = \frac{1}{\omega}$

$\Rightarrow z = \infty$ at $\omega = 0$.

So, $z = \infty$ is non-isolated essential singularity.